

ONLINE MASTERS IN DATA SCIENCE

DSC 257R - UNSUPERVISED LEARNING

THE AUTOENCODER FORMALISM

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Unnamed Structure

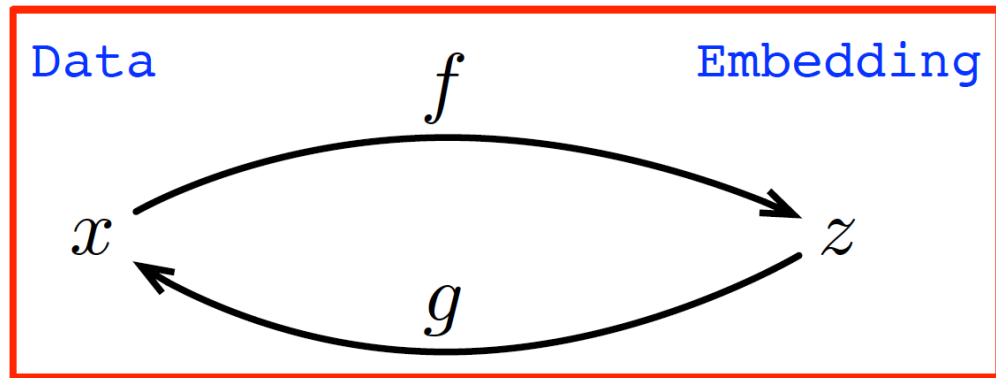
So far: representations that capture various types of structure:

- clusters
- linear subspaces
- manifolds
- explicit distance and similarity information

Can we find representations that capture:

- **multiple types of structure**, simultaneously
- **other structure for which we don't even have definitions?**

Deterministic Autoencoders



- f : encoding map
- g : decoding map

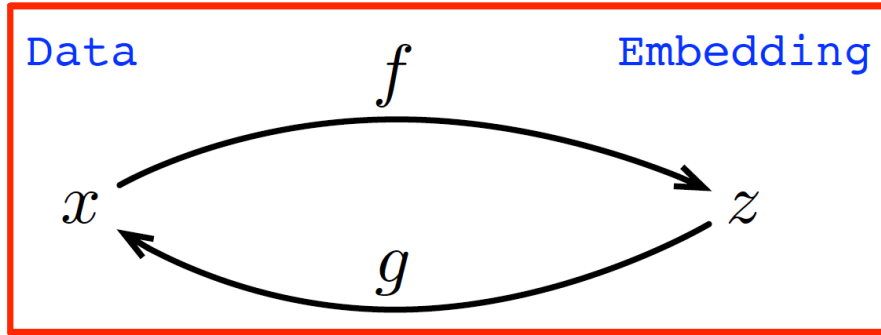
z should be *simpler* than x :

- e.g., lower-dimensional or sparser
- expose key structure in x

Typically: find f, g to minimize reconstruction loss $\mathbb{E}_X [\ell(X, g(f(X)))]$

PCA as an Autoencoder

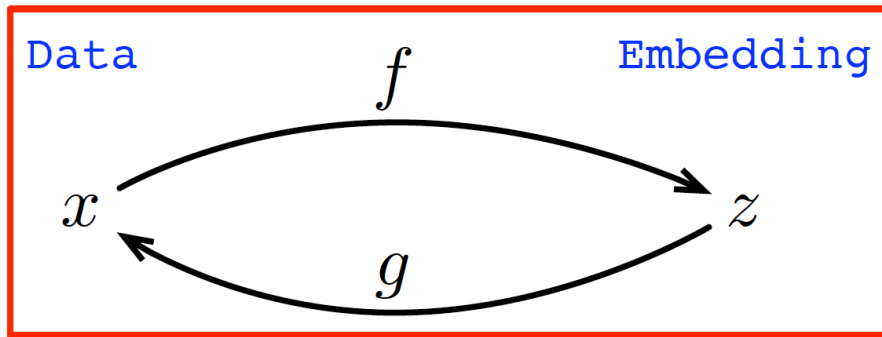
Use PCA to map data from $\mathbb{R}^d$ to $\mathbb{R}^k$.



- What are f and g ?
- What loss function is being minimized?

K -means as an Autoencoder

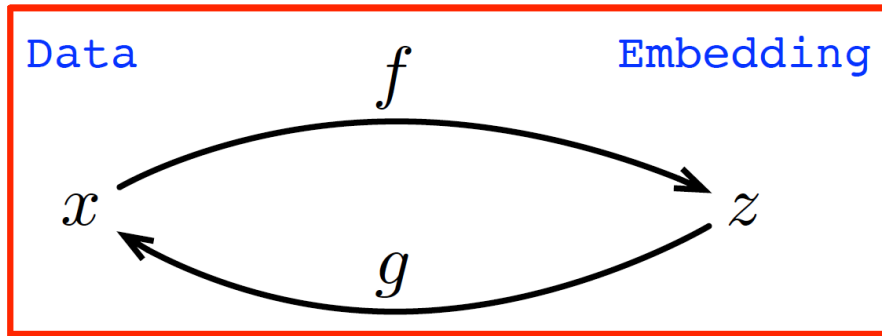
Apply k -means clustering to data in $\mathbb{R}^d$.



- What are f and g ?
- What loss function is being minimized?

More General Autoencoder

Allow f, g to be arbitrary functions, e.g., neural nets.



- f : encoding map
- g : decoding map

Here the embedding is often constrained to be **simple** in some way, e.g.:

- Lower-dimensional than x
- Sparse